COURSE

Pearson Edexcel International GCSE in Business (4BS1)

Working with suppliers

Section 5 · 5.2 Production · 5.2.2 Principles of lean production (just-in-time)

Revision for Section 5: each firm works it out on its own deals.

Class	Topic
Pearson Edexcel International GCSE Business (4BS1)	**Working with suppliers**

THE LESSON

Setup					2 min
Round 1	Decisions 3 min · Deals 5 min · Topic tasks 4 min				
Firm decisions	**Deals**	TASK 1 **Stock holding cost**	TASK 2 **Too much stock**		
Round 2	Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 2 min				
Firm decisions	**Deals**	TASK 3 **Rank supplier cost**	TASK 4 **Saving from JIT**	**B2B deal**	
Round 3	Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 2 min				
Firm decisions	**Deals**	TASK 5 **JIT or hold stock**	TASK 6 **JIT vs discount**	**B2B deal**	
Final questions					3 min

45-minute setting shown. The 30-minute setting has one topic task per round (Tasks 3, 5 and 6) and one B2B deal. 60 and 75 add more deals.

Each step

Firm decisions	Each role makes its decision on its own page.
Deals	Each pupil trades with a classmate in another firm.
Topic task	The firm answers together. Everyone presses Agree.
B2B deal	One firm sells a lot of cars to another firm.
Final questions	Each pupil answers six questions.

Firm decisions

CEO	**Staff**	Hire staff. Each costs a salary and gives one benefit.
	Sabotage FROM ROUND 2	Hit a rival firm's cash, at a cost. A Legal Counsel blocks it.
Sales & Marketing	**Marketing**	Book one advert campaign to lift the next sales.
Finance	**Capital & Loans**	Borrow cash. The firm pays it back with interest.
	Investments	Buy property or funds. The value can rise or fall.
Operations	**Buying**	Hire the Procurement Specialist or the Stock Controller for the round.

B2B deal · Rounds 2 and 3

One firm sells a lot of 3 cars to another firm (4 in Round 3). With an odd number of firms, two firms offer to sell and the lower price wins.

Topic tasks

Each firm gets its own figures. The method is the same for every firm.

Task 1

ROUND 1

Cars wait in your yard.

Cost of holding the cars?

WORK OUT THE NUMBER · EXAMPLE FIGURES

You hold for them	**8 cars**
Holding cost per car a week	**£100**
Weeks you hold them	**4 weeks**

METHOD

Holding cost = cars × cost per car a week × weeks

8 × £100 × 4 = £3,200

Why: Stock that waits costs £800 every week.

Task 2

ROUND 1

Your yard holds far more cars than you sell in a month.

Which two problems are most likely?

PICK TWO

- ✔ **Cash tied up in stock**
- ✔ **Higher storage costs**
- ○ Lost sales from empty space
- ○ Lower insurance costs
- ○ Running out of cars to sell

Why: Extra stock holds cash and costs money to keep.

Task 3

ROUND 2

Three suppliers quote for 10 cars, delivered.

Order by total cost, cheapest first?

PUT THESE IN ORDER, FIRST TO LAST · RIGHT ORDER

1. **£8,100 a car, £1,200 delivery**
2. **£8,000 a car, £2,500 delivery**
3. **£8,300 a car, £300 delivery**

Why: £82,200, £82,500 and £83,300: the lowest price per car is not the cheapest order.

Task 4

ROUND 2

Just in time, or in case?

Saving a month from JIT?

WORK OUT THE NUMBER, THEN PICK IT · EXAMPLE FIGURES

Spare cars held, just in case	**7 cars**
Holding cost per car a week	**£70**
Extra JIT deliveries, a week	**£200**
Weeks in the month	**4 weeks**

METHOD

Saving = weeks × (spare cars × holding cost − extra deliveries)

4 × (7 × £70 − £200) = £1,160

Why: JIT saves the holding cost, but the extra deliveries cost £800 a month.

Task 5

ROUND 3

Your supplier is two hours away and always on time. Hold 40 cars, or hold 10 and reorder often?

Which should you choose, and why?

PICK A OR B AND GIVE A REASON

- ✔ **Hold 10 and reorder often**
- ○ Hold 40 cars

Why: A reliable, nearby supplier lets you hold less and free up cash.

Task 6

ROUND 3

JIT loses a bulk discount.

Gain a month from JIT? Minus allowed.

WORK OUT THE NUMBER · EXAMPLE FIGURES

Cars bought a month	**13 cars**
Price per car	**£8,000**
Bulk discount lost with JIT	**3%**
Holding cost saved a month	**£2,000**

METHOD

Gain = holding cost saved – cars × price × discount lost

13 × £8,000 × 3% = £3,120; £2,000 − £3,120 = −£1,120

Why: Smaller, frequent orders save holding costs but can lose bulk-buying discounts.

Final questions

6
questions for each pupil

159
Level 2 Business questions

10
of them on working with suppliers

The game draws 6 questions on this topic.

1 What is one risk of relying on a single overseas parts supplier?
A problem abroad could stop its supply.

2 Which evidence set is most useful when selecting a supplier for regularly needed vehicle parts?
Price, quality, reliability and delivery performance.

3 What is an opportunity cost of holding too much vehicle stock?
The cash cannot be used for anything else.

4 Supplier A is slightly cheaper, but frequently delivers parts late. Supplier B is reliable. Which factor supports choosing B for urgent repairs?
Dependability of delivery.

5 A workshop keeps almost no spare parts and orders only when needed. What is the main risk of this just-in-time approach?
A late delivery can stop work.

6 Delivery fuel costs rise sharply. Which operational response is most likely to reduce the impact?
Plan shorter routes with fuller vans.

7 A key supplier enters insolvency. Which information is most important first?
Open orders, stock cover and backup suppliers.

Final questions

8 High-value parts repeatedly disappear from stock records. Which control system is most useful?

Restrict access, log movements and reconcile counts with records.

9 A supply chain is every business that helps get a product to the customer. A tyre maker's factory floods. Using this, what happens to a workshop?

Its tyre deliveries may stop, though it did nothing wrong.

10 A reliable supplier delivers the right parts, in full, when promised. The cheapest supplier delivers late one time in three. What does staying with it really cost?

Delayed jobs and waiting customers can cost more than the saving.